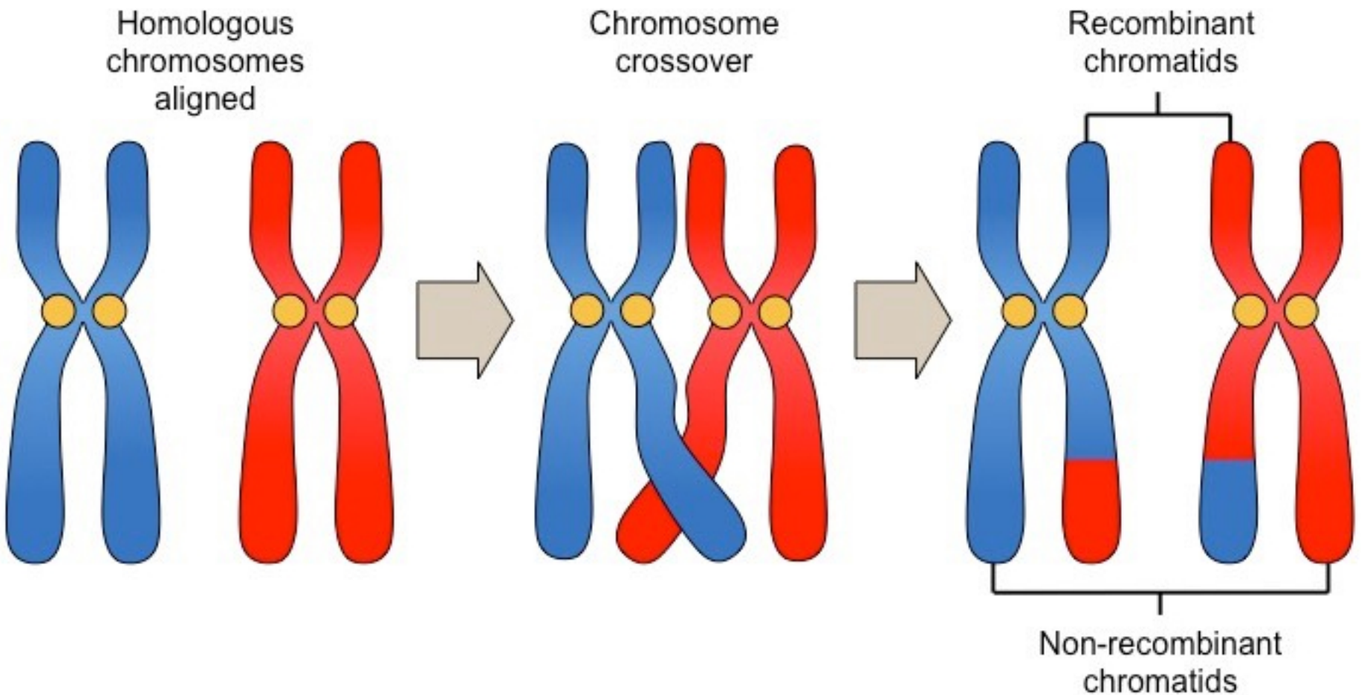
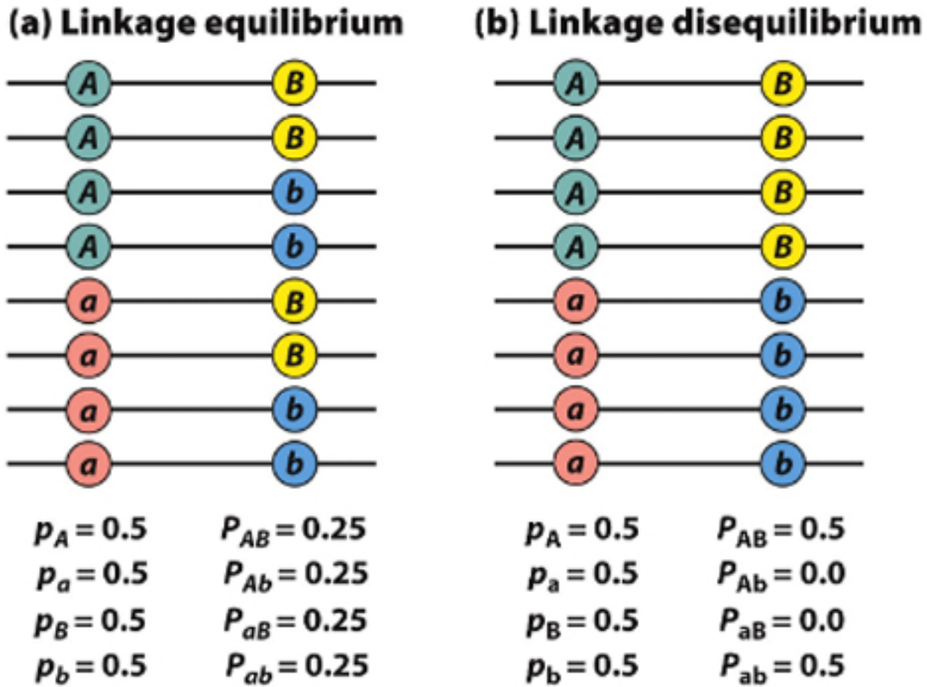

Homologous chromosomes aligned
Chromosome crossover
Recombinant chromatids
Non-recombinant chromatids


(a) Linkage equilibrium
A A A A a a a a
B B b b B B b b
$p_A = 0.5$ $P_{AB} = 0.25$
$p_a = 0.5$ $P_{Ab} = 0.25$
$p_B = 0.5$ $P_{aB} = 0.25$
$p_b = 0.5$ $P_{ab} = 0.25$
(b) Linkage disequilibrium
A A A A a a a a
B B B B b b b b
$p_A = 0.5$ $P_{AB} = 0.5$
$p_a = 0.5$ $P_{Ab} = 0.0$
$p_B = 0.5$ $P_{aB} = 0.0$
$p_b = 0.5$ $P_{ab} = 0.5$